

BRAF fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes BRAF gene fusions from the msk_impact_50k_2026 study, covering 179 fusion events. 151/179 fusions (84.4%) were in-frame. 163/179 fusions (91.1%) retained the PF07714 domain. Domain retention was tested with Fisher's exact test ($p=0.0133676$, statistically significant at $\alpha=0.05$). Compared to the literature benchmark (PMC5461196: 100.0% in-frame, 100.0% domain-retained), this run observed 84.4% in-frame and 91.1% domain retention.

1 Results summary

stats analysis was skipped.

1.1 Fusion partner frequency

Fusion-partner frequency was tabulated across 179 analyzed fusion events, identifying 67 distinct fusion partner genes. The most frequent partner was KIAA1549, observed in 43 events.

1.6 Composite evidence score

Composite evidence score reported: confidence_certainty sub-score excluded: no applicable confidence_stats MLE/CI result was supplied.

1.2 Domain retention

PF07714 domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 142/151 (94.0%) of in-frame fusions retained the domain, $p=0.0133676$ (statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.045954.

1.7 Exon retention

Exon retention reported: No target exon was supplied in params or GeneConfig.

1.3 Domain disruption

Disruption of the configured disruption-required domain(s) was tested with Fisher's exact test comparing in-frame fusions against all others; 142/151 (94.0%) of in-frame fusions disrupted the domain, $p=0.041705$ (statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 1.

1.8 Joint partner

Joint partner reported: BRAF has no gene_pair configured; joint-partner analysis was skipped.

1.4 Cutpoint detection

Cutpoint detection scanned 178 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 327 aa (permutation-corrected $p=0.019802$, statistically significant at $\alpha=0.05$). This is 47 aa from the nearest configured domain boundary at 280 aa.

1.5 Corroborating confidence statistics

Corroborating confidence statistics were not computed for this run: BRAF has no group_field configured; confidence-

2 Tables

2.1 composite_score composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty	Components_applicable	Composite_score	Rank
AGK	14	0.0782122905027933	0.13376762457977448	0.0	0.9846938775510204		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.22734408325381628	1
ABCC2	1	0.00558659217877095	0.13376762457977448	0.0	1.0		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.20568653755397212	2
ITGB5	1	0.00558659217877095	0.13376762457977448	0.0	1.0		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.20568653755397212	3
MKLN1	1	0.00558659217877095	0.13376762457977448	0.0	1.0		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.20568653755397212	4
NEO1	1	0.00558659217877095	0.13376762457977448	0.0	1.0		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.20568653755397212	5
PWWP2A	1	0.00558659217877095	0.13376762457977448	0.0	1.0		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.20568653755397212	6
TBXAS1	1	0.00558659217877095	0.13376762457977448	0.0	1.0		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.20568653755397212	7
UBE2H	1	0.00558659217877095	0.13376762457977448	0.0	1.0		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.20568653755397212	8
WEE2	1	0.00558659217877095	0.13376762457977448	0.0	1.0		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.20568653755397212	9
KIAA1549	43	0.24022346368715083	0.13376762457977448	0.0	0.436046511627907		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18990657999474986	10
CLIP1	2	0.0111731843575419	0.13376762457977448	0.0	0.7589285714285714		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.16737016351832432	11
LUC7L2	1	0.00558659217877095	0.13376762457977448	0.0	0.6428571428571428		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1461627280301626	12
CUL1	5	0.027932960893854747	0.13376762457977448	0.0	0.5249999999999999		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.13396866045900005	13
SND1	17	0.09497206703910614	0.13376762457977448	0.0	0.3897058823529411		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.13376600956624068	14
FAM131B	3	0.01675977653631285	0.13376762457977448	0.0	0.5178571428571428		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.12905378948267654	15
CARMIL1	2	0.0111731843575419	0.13376762457977448	0.0	0.5267857142857143		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.12867968732784812	16
MAD1L1	2	0.0111731843575419	0.13376762457977448	0.0	0.5267857142857143		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.12867968732784812	17
AKAP9	2	0.0111731843575419	0.13376762457977448	0.0	0.5223214285714286		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1279356397088005	18

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty	Components_applicable	Composite_score	Rank
EXOC4	2	0.0111731843575419	0.13376762457977448	0.0	0.5178571428571428		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1271915920897529	19
ABCC1	1	0.00558659217877095	0.13376762457977448	0.0	0.5267857142857143		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.12681748993492448	20
CAPZA2	1	0.00558659217877095	0.13376762457977448	0.0	0.5267857142857143		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.12681748993492448	21
KLHL12	1	0.00558659217877095	0.13376762457977448	0.0	0.5267857142857143		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.12681748993492448	22
LRBA	1	0.00558659217877095	0.13376762457977448	0.0	0.5267857142857143		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.12681748993492448	23
OSBPL7	1	0.00558659217877095	0.13376762457977448	0.0	0.5267857142857143		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.12681748993492448	24
PRIM2	1	0.00558659217877095	0.13376762457977448	0.0	0.5267857142857143		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.12681748993492448	25
PRKAR1B	1	0.00558659217877095	0.13376762457977448	0.0	0.5267857142857143		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.12681748993492448	26
PTPRZ1	1	0.00558659217877095	0.13376762457977448	0.0	0.5267857142857143		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.12681748993492448	27
RGS3	1	0.00558659217877095	0.13376762457977448	0.0	0.5267857142857143		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.12681748993492448	28
TRA2B	1	0.00558659217877095	0.13376762457977448	0.0	0.5267857142857143		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.12681748993492448	29
TRIM24	8	0.0446927374301676	0.13376762457977448	0.0	0.4419642857142857		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.12571596692348527	30
CCDC6	1	0.00558659217877095	0.13376762457977448	0.0	0.5178571428571428		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.12532939469682924	31
CMAS	1	0.00558659217877095	0.13376762457977448	0.0	0.5178571428571428		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.12532939469682924	32
ERC1	1	0.00558659217877095	0.13376762457977448	0.0	0.5178571428571428		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.12532939469682924	33
MINDY4	1	0.00558659217877095	0.13376762457977448	0.0	0.5178571428571428		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.12532939469682924	34
PHTF2	1	0.00558659217877095	0.13376762457977448	0.0	0.5178571428571428		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.12532939469682924	35
SNX8	1	0.00558659217877095	0.13376762457977448	0.0	0.5178571428571428		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.12532939469682924	36
AGAP3	6	0.0335195530726257	0.13376762457977448	0.0	0.4196428571428571		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.11827133404239987	37
NRF1	2	0.0111731843575419	0.13376762457977448	0.0	0.4642857142857143		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.11826302066118148	38
GTF2IRD1	3	0.01675977653631285	0.13376762457977448	0.0	0.4107142857142857		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.11119664662553369	39

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty	Components_applicable	Composite_score	Rank
TMPRSS2	4	0.0223463687150838	0.13376762457977448	0.0	0.37797619047619047		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.10760249481210814	40
ATF7	1	0.00558659217877095	0.13376762457977448	0.0	0.4107142857142857		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1074722518396864	41
CLEC2L	1	0.00558659217877095	0.13376762457977448	0.0	0.4107142857142857		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1074722518396864	42
DBI	1	0.00558659217877095	0.13376762457977448	0.0	0.4107142857142857		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1074722518396864	43
DOCK4	1	0.00558659217877095	0.13376762457977448	0.0	0.4107142857142857		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1074722518396864	44
GIPC2	1	0.00558659217877095	0.13376762457977448	0.0	0.4107142857142857		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1074722518396864	45
GTF2I	1	0.00558659217877095	0.13376762457977448	0.0	0.4107142857142857		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1074722518396864	46
KCND2	1	0.00558659217877095	0.13376762457977448	0.0	0.4107142857142857		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1074722518396864	47
NFIA	1	0.00558659217877095	0.13376762457977448	0.0	0.4107142857142857		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1074722518396864	48
OSBPL9	1	0.00558659217877095	0.13376762457977448	0.0	0.4107142857142857		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1074722518396864	49
PAK2	1	0.00558659217877095	0.13376762457977448	0.0	0.4107142857142857		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1074722518396864	50
PRKAR2B	1	0.00558659217877095	0.13376762457977448	0.0	0.4107142857142857		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1074722518396864	51
RRBP1	1	0.00558659217877095	0.13376762457977448	0.0	0.4107142857142857		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1074722518396864	52
SETBP1	1	0.00558659217877095	0.13376762457977448	0.0	0.4107142857142857		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1074722518396864	53
SLC45A3	1	0.00558659217877095	0.13376762457977448	0.0	0.4107142857142857		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1074722518396864	54
SPOCK1	1	0.00558659217877095	0.13376762457977448	0.0	0.4107142857142857		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1074722518396864	55
ZC3HAV1	1	0.00558659217877095	0.13376762457977448	0.0	0.4107142857142857		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1074722518396864	56
ZNF207	1	0.00558659217877095	0.13376762457977448	0.0	0.4107142857142857		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1074722518396864	57
CDK5RAP2	4	0.0223463687150838	0.13376762457977448	0.0	0.3660714285714286		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.10561836782798116	58
MKRN1	10	0.055865921787709494	0.13376762457977448	0.0	0.0625		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.06619631409028497	59
EPB41	1	0.00558659217877095	0.13376762457977448	0.0	0.008928571428571397		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.04050796612540068	60

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty	Components_applicable	Composite_score	Rank
PJA2	1	0.00558659217877095	0.13376762457977448	0.0	0.008928571428571397		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.04050796612540068	61
RBM33	1	0.00558659217877095	0.13376762457977448	0.0	0.008928571428571397		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.04050796612540068	62
RBMS3	1	0.00558659217877095	0.13376762457977448	0.0	0.008928571428571397		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.04050796612540068	63
PLPP1	1	0.00558659217877095	0.13376762457977448	0.0	0.0		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.03901987088730545	64
SCRIB	1	0.00558659217877095	0.13376762457977448	0.0	0.0		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.03901987088730545	65
SVOPL	1	0.00558659217877095	0.13376762457977448	0.0	0.0		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.03901987088730545	66
TRIM33	1	0.00558659217877095	0.13376762457977448	0.0	0.0		['cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.03901987088730545	67

2.2 cutpoint_detection cutpoint_scan

CT

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
238	1	162	0	15		1.0	-0.0
287	1	162	1	14	0.08641975308641975	0.16187392877547135	0.7908230925727642
303	2	161	1	14	0.17391304347826086	0.23330512802756184	0.6320757153442778
327	19	144	6	9	0.19791666666666666	0.00883450422715351	2.053817817119616
380	68	95	8	7	0.6263157894736842	0.422537724516002	0.37413451081430776
381	114	49	9	6	1.5510204081632653	0.5596655980047007	0.2520713878510084
393	140	23	12	3	1.5217391304347827	0.46313499579777323	0.33429240123968906
400	141	22	12	3	1.6022727272727273	0.44662922460047316	0.3500528628421283
438	156	7	13	2	3.4285714285714284	0.16897876327149897	0.7721678727445281
439	163	0	14	1		0.08426966292134831	1.0743287432532127

2.3 domain_disruption frame_domain_contingency_table

142	22
9	5

2.4 domain_disruption domain_disruption_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained
ras_binding	RAS-binding domain	178	14	0	164	0.0
cysteine_rich	Cysteine-rich domain	178	14	1	163	0.005699893955461294

2.5 domain_retention frame_domain_contingency_table

142	21
9	6

2.6 domain_retention domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained
kinase	Protein kinase domain	178	163	0	15	0.0

2.7 frequency Partner_gene_counts

Partner_gene	Event_count
ABCC1	1
ABCC2	1
AGAP3	6
AGK	14
AKAP9	2
ATF7	1
CAPZA2	1
CARMIL1	2
CCDC6	1
CDK5RAP2	4
CLEC2L	1
CLIP1	2
CMAS	1
CUL1	5
DBI	1
DOCK4	1
EPB41	1
ERC1	1
EXOC4	2
FAM131B	3
GIPC2	1
GTF2I	1
GTF2IRD1	3
ITGB5	1
KCND2	1
KIAA1549	43
KLHL12	1
LRBA	1
LUC7L2	1
MAD1L1	2
MINDY4	1
MKLN1	1
MKRN1	10

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2.8 Per-event results (results.tsv)

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event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_pos	domain_pos	main_ref	conf_fra	unintended_domains	disrupted_domains	annotation	effect_on_fra
EVT-P-0050574-T02-IM6-4	P-0050574-T02-IM6		AGAP3-BRAF	AGAP3	in-frame	three_prime	140493208	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain	AGAP3 (NM_031946) - BRAF (NM_004333) fusion: c.1222-2059:AGAP3_c.1140+900:BRAFInv	
EVT-P-0039799-T01-IM6-5	P-0039799-T01-IM6		AGAP3-BRAF	AGAP3	in-frame	three_prime	140482317	10	438	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain	AGAP3 (NM_031946) - BRAF (NM_004333) fusion: c.1221+416:AGAP3_c.1314+504:BRAFInv	
EVT-P-0050931-T01-IM6-6	P-0050931-T01-IM6		AGK-BRAF	AGK	in-frame	three_prime	140499251	7	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain	AGK (NM_018238) - BRAF (NM_004333) fusion: c.102-14294:AGK_c.980+911:BRAFInv	
EVT-P-0042835-T03-IM6-7	P-0042835-T03-IM6		AGK-BRAF	AGK	in-frame	three_prime	140497995	7	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain	AGK (NM_018238) - BRAF (NM_004333) fusion: c.102-13644:AGK_c.980+2167:BRAFInv	
EVT-P-0042835-T02-IM6-8	P-0042835-T02-IM6		AGK-BRAF	AGK	in-frame	three_prime	140497995	7	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain	AGK (NM_018238) - BRAF (NM_004333) fusion: c.102-13644:AGK_c.980+2167:BRAFInv	
EVT-P-0050635-T01-IM6-9	P-0050635-T01-IM6		AKAP9-BRAF	AKAP9	in-frame	three_prime	140487501	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain	AKAP9 (NM_005751) - BRAF (NM_004333) fusion: c.6613-1489:AKAP9_c.1141-117:BRAFInv	
EVT-P-0030237-T03-IM6-10	P-0030237-T03-IM6		AKAP9-BRAF	AKAP9	in-frame	three_prime	140492357	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain	AKAP9 (NM_005751) - BRAF (NM_004333) fusion: c.3318+1378:AKAP9_c.1140+1751:BRAFInv	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	main_ref	disrupted_1	disrupted_2	disrupted_3	disrupted_4	description	annotation	effect_on_frame
EVT-P-0023821-T01-IM6-12	P-0023821-T01-IM6		ATF7-BRAF	ATF7	in-frame	three_prime	140483233	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			ATF7 (NM_006856) - BRAF (NM_004333) Rearrangement: t(7;12)(q34;q13.13)(chr7:g.140483233::c.53913299) Protein Fusion: in frame {ATF7:BRAF}	NA	
EVT-P-0007861-T01-IM5-13	P-0007861-T01-IM5		BRAF-AGAP3	AGAP3	in-frame	three_prime	140490697	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			AGAP3 (NM_031946) - BRAF (NM_004333) Rearrangement: c.1326+220:AGAP3_c.1141-3313:BRAFinv Protein fusion: in frame (AGAP3-BRAF)	NA	
EVT-P-0021519-T02-IM6-14	P-0021519-T02-IM6		BRAF-AGAP3	AGAP3	in-frame	three_prime	140493112	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			AGAP3 (NM_031946) - BRAF (NM_004333) Rearrangement: c.1326+142:AGAP3_c.1140+996:BRAFinv Protein fusion: in frame (AGAP3-BRAF)	NA	
EVT-P-0009757-T01-IM5-15	P-0009757-T01-IM5		BRAF-AGAP3	AGAP3	in-frame	three_prime	140493871	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			AGAP3 (NM_031946) - BRAF (NM_004333) Rearrangement: c.1222-2154:AGAP3_c.1140+287:BRAFinv Protein fusion: in frame (AGAP3-BRAF)	NA	
EVT-P-0036886-T01-IM6-16	P-0036886-T01-IM6		BRAF-AGK	AGK	in-frame	three_prime	140494309	8	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			AGK (NM_018238) - BRAF (NM_004333) Rearrangement: c.101+10859:AGK_c.981-42:BRAFinv Protein Fusion: in frame {AGK:BRAF}	NA	
EVT-P-0025561-T03-IM6-17	P-0025561-T03-IM6		BRAF-AGK	AGK	in-frame	three_prime	140498719	7	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			AGK (NM_018238) - BRAF (NM_004333) fusion (AGK exons 1-2 fused to BRAF exons 8-18): c.980+1288:BRAF_c.101+772:AGKinv Protein Fusion: in frame {AGK:BRAF}	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	main_ref	clonally_deleted	domains	disrupted_domains	description	annotation	effect_on_fra
EVT-P-0000913-T02-IM5-23	P-0000913-T02-IM5		BRAF-AGK	AGK	in-frame	three_prime	140496386	8	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) Fusion (AGK exon 2 fused with BRAF exon 8) : c.101+13932:AGK_c.981-2119:BRAFInv Protein fusion: in frame (AGK-BRAF)	NA	
EVT-P-0029531-T01-IM6-24	P-0029531-T01-IM6		BRAF-AGK	AGK	unknown	three_prime	140500233	7	303	False	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) fusion (AGK exons 1-2 fused to BRAF exons 7-18): c.101+4572:AGK_c.909:BRAFInv Protein Fusion: mid-exon {AGK:BRAF} AGK (NM_018238) - BRAF (NM_004333) fusion: c.101+8348:AGK_c.980+2629:BRAFInv Protein Fusion: in frame {AGK:BRAF} AGK (NM_018238) - BRAF (NM_004333) rearrangement : c.101+12646:AGK_c.980+2552:BRAFInv Protein fusion: in frame (AGK-BRAF)	NA	
EVT-P-0030746-T01-IM6-25	P-0030746-T01-IM6		BRAF-AGK	AGK	in-frame	three_prime	140497533	7	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) fusion: c.101+8348:AGK_c.980+2629:BRAFInv Protein Fusion: in frame {AGK:BRAF} AGK (NM_018238) - BRAF (NM_004333) rearrangement : c.101+12646:AGK_c.980+2552:BRAFInv Protein fusion: in frame (AGK-BRAF)	NA	
EVT-P-0010945-T01-IM5-26	P-0010945-T01-IM5		BRAF-AGK	AGK	in-frame	three_prime	140497610	7	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) rearrangement : c.101+12646:AGK_c.980+2552:BRAFInv Protein fusion: in frame (AGK-BRAF)	NA	
EVT-P-0049035-T01-IM6-85	P-0049035-T01-IM6		BRAF-CLEC2L	CLEC2L	in-frame	five_prime	140485799	9	393	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - CLEC2L (NM_001080511) rearrangement: c.1177+1549:BRAF_c.191-5932:CLEC2LInv Protein Fusion: in frame {BRAF:CLEC2L} CUL1(NM_0002592) - BRAF (NM_004333) Rearrangement :c.790-427:CUL1_c.1140+297:BRAFInv Protein fusion: in frame (CUL1-BRAF)	NA	
EVT-P-0004795-T01-IM5-87	P-0004795-T01-IM5		BRAF-CUL1	CUL1	in-frame	three_prime	140493811	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		BRAF (NM_004333) - CUL1(NM_0002592) - BRAF (NM_004333) Rearrangement :c.790-427:CUL1_c.1140+297:BRAFInv Protein fusion: in frame (CUL1-BRAF)	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	main_ref	donor_frame	inframe	domains	disrupted_domains	description	annotation	effect_on_fra
EVT-P-0024995-T01-IM6-121	P-0024995-T01-IM6		CMAS-BRAF	CMAS	unknown	three_prime	140488046	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			BRAF (NM_004333) rearrangement: t(7;12)(q34;p12.1)(chr7:g.140488046::chr12:g.1140+1215:BRAF)in	NA	
EVT-P-0039648-T01-IM6-122	P-0039648-T01-IM6		CUL1-BRAF	CUL1	in-frame	three_prime	140492893	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			Fusion: {CMAS:BRAF} CUL1 (NM_003592) - BRAF (NM_004333) fusion: c.789+2950;CUL1_c.1140+1215:BRAFin	NA	
EVT-P-0039648-T03-IM6-123	P-0039648-T03-IM6		CUL1-BRAF	CUL1	in-frame	three_prime	140492893	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			Fusion: {CUL1:BRAF} CUL1 (NM_003592) - BRAF (NM_004333) fusion: c.789+2950;CUL1_c.1140+1215:BRAFin	NA	
EVT-P-0064088-T01-IM7-124	P-0064088-T01-IM7		CUL1-BRAF	CUL1	in-frame	three_prime	140489451	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			Fusion: {CUL1:BRAF} CUL1 (NM_003592) - BRAF (NM_004333) fusion: c.789+1956;CUL1_c.1141-2067:BRAFinv	NA	
EVT-P-0068173-T01-IM7-125	P-0068173-T01-IM7		CUL1-BRAF	CUL1	in-frame	three_prime	140491990	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			Fusion: {CUL1:BRAF} CUL1 (NM_003592) - BRAF (NM_004333) fusion: c.789+191;CUL1_c.1140+2118:BRAFin	NA	
EVT-P-0050196-T01-IM6-126	P-0050196-T01-IM6		DBI-BRAF	DBI	in-frame	three_prime	140486284	9	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			Fusion: {CUL1:BRAF} DBI (NM_001178017) - BRAF (NM_004333) fusion: t(2;7)(q14.2;q34)(chr2:g.120128894::chr7:g.1140+1215:BRAF)in	NA	
EVT-P-0064588-T01-IM7-127	P-0064588-T01-IM7		DOCK4-BRAF	DOCK4	out-of-frame	three_prime	140483872	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			Fusion: {DBI:BRAF} DOCK4 (NM_014705) - BRAF (NM_004333) fusion: c.4650+298;DOCK4_c.1178-915:BRAFdup	NA	
EVT-P-0014833-T02-IM6-128	P-0014833-T02-IM6		ERC1-BRAF	ERC1	in-frame	three_prime	140490668	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			Fusion: {DOCK4:BRAF} ERC1 (NM_178040) - BRAF (NM_004333) fusion: t(7;12)(q34;p13.33)(chr7:g.140490668::chr12:g.1140+1215:BRAF)in	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_pos	breakpoint_ref	breakpoint_chrom	protein_position	protein_domain	main_retention	donor_insertion	transcript_maintained	disrupted_domains	domains_annotation	station_site	effect_on_fra
EVT-P-0036937-T01-IM6-129	P-0036937-T01-IM6		EXOC4-BRAF	EXOC4	out-of-frame	three_prime	140490241	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich do-main	EXOC4 (NM_021807) - BRAF (NM_004333) fusion: c.1514+93469:EXOC4_c.1141-2857:BRAFInv Protein Fusion: out of frame {EXOC4:BRAF}	NA (NM_021807)	
EVT-P-0024206-T01-IM6-130	P-0024206-T01-IM6		EXOC4-BRAF	EXOC4	out-of-frame	three_prime	140487956	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich do-main	EXOC4 (NM_021807) - BRAF (NM_004333) rearrangement: c.87-8072:EXOC4_c.1141-572:BRAFInv Protein Fusion: out of frame {EXOC4:BRAF}	NA (NM_021807)	
EVT-P-0047918-T04-IM7-131	P-0047918-T04-IM7		FAM131B-BRAF	FAM131B	in-frame	three_prime	140487437	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich do-main	FAM131B (NM_001031690) - BRAF (NM_004333) fusion: c.175-20:FAM131B_c.1141-53:BRAFdel Protein Fusion: in frame {FAM131B:BRAF}	NA (NM_001031690)	
EVT-P-0047918-T02-IM6-132	P-0047918-T02-IM6		FAM131B-BRAF	FAM131B	in-frame	three_prime	140487437	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich do-main	FAM131B (NM_001031690) - BRAF (NM_004333) fusion: c.175-20:FAM131B_c.1141-53:BRAFdel Protein Fusion: in frame {FAM131B:BRAF}	NA (NM_001031690)	
EVT-P-0044845-T01-IM6-134	P-0044845-T01-IM6		GTF2I-BRAF	GTF2I	unknown	three_prime	140485631	9	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich do-main	GTF2I (NM_032999) - BRAF (NM_004333) fusion: c.768:GTF2I_c.1177+1717:BRAFInv Protein Fusion: mid-exon {GTF2I:BRAF}	NA (NM_032999)	
EVT-P-0000828-T05-IM6-135	P-0000828-T05-IM6		GTF2IRD1-BRAF	GTF2IRD1	in-frame	three_prime	140483546	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich do-main	GTF2IRD1 (NM_005685) - BRAF (NM_004333) rearrangement: c.2320+14147:GTF2IRD1_c.1178-589:BRAFInv Protein Fusion: in frame {GTF2IRD1:BRAF}	NA (NM_005685)	
EVT-P-0046745-T01-IM6-136	P-0046745-T01-IM6		KCND2-BRAF	KCND2	out-of-frame	three_prime	140485867	9	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich do-main	KCND2 (NM_012281) - BRAF (NM_004333) fusion: c.1115+91797:KCND2_c.1177+1481:BR		

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	is_main_ref	is_in_frame	is_retained	disrupted_domain	disrupted_domain	description_annotation	annotation_start	effect_on_fusion
EVT-P-0056952-T01-IM6-138	P-0056952-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140493821	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5248-141:KIAA1549_c.1140+287:BRAFdup		
EVT-P-0044270-T02-IM6-139	P-0044270-T02-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140491645	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5247+1560:KIAA1549_c.1140+2463:BRAFdup		
EVT-P-0027592-T01-IM6-140	P-0027592-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140492343	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused with BRAF exons 9-18) : c.5247+769:KIAA1549_c.1140+1765:BRAFdup		
EVT-P-0055099-T03-IM6-141	P-0055099-T03-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140490663	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4929+1357:KIAA1549_c.1141-3279:BRAFdup		
EVT-P-0058584-T01-IM7-142	P-0058584-T01-IM7		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140491493	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5247+1668:KIAA1549_c.1140+2615:BRAFdup		
EVT-P-0066030-T01-IM7-143	P-0066030-T01-IM7		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140482785	10	438	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5248-1172:KIAA1549_c.1314+36:BRAFdup		
EVT-P-0035153-T01-IM6-144	P-0035153-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140491093	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			KIAA1549 (NM_001164665) - BRAF (NM_004333) Fusion (KIAA1549 exon15 fused with BRAF exon9) : c.4930-2046:KIAA1549_c.1140+3015:BRAFdup		

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint	breakpoint	breakpoint	protein	protein	breakpoint	main_ref	clonotype	variant	disrupted_domain	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0035153-T03-IM6-145	P-0035153-T03-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140491093	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4930-2046:KIAA1549_c.1140-3015:BRAFdup		
EVT-P-0049925-T01-IM6-146	P-0049925-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140489497	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5247+301:KIAA1549_c.1141-2113:BRAFdup		
EVT-P-0035844-T03-IM7-147	P-0035844-T03-IM7		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140493285	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4033-1202:KIAA1549_c.1140-823:BRAFdup		
EVT-P-0035844-T04-IM7-148	P-0035844-T04-IM7		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140493285	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4033-1202:KIAA1549_c.1140-823:BRAFdup		
EVT-P-0017693-T02-IM6-149	P-0017693-T02-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140489537	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4930-495:KIAA1549_c.1141-2153:BRAFdup		
EVT-P-0062963-T01-IM7-150	P-0062963-T01-IM7		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140482198	10	438	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4930-27:KIAA1549_c.1314+623:BRAFdup		
EVT-P-0012575-T01-IM5-151	P-0012575-T01-IM5		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140490295	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-15 with BRAF exons 9-18): c.4929+793:KIAA1549_c.1141-2911:BRAFdup		

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	is_main_ref	is_in_frame	is_retained	disrupted_domain	disrupted_domain	description_annotation	annotation_start	effect_on_fra
EVT-P-0052411-T05-IM7-152	P-0052411-T05-IM7		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140493530	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4930-2296:KIAA1549_c.1140-578:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}		
EVT-P-0034117-T01-IM6-153	P-0034117-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140483812	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4346-57:KIAA1549_c.1178-855:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}		
EVT-P-0052411-T03-IM6-154	P-0052411-T03-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140493530	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4930-2296:KIAA1549_c.1140-578:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}		
EVT-P-0009097-T01-IM5-155	P-0009097-T01-IM5		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140488817	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused in frame with BRAF exons 9-18, which include the BRAF kinase domain): c.5247+3837:KIAA1549_c.1141-1433:BRAFdup Protein Fusion: in frame (KIAA1549-BRAF)		
EVT-P-0005121-T02-IM5-156	P-0005121-T02-IM5		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140491248	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-15 with BRAF exons 9-18): c.4929+2454:KIAA1549_c.1140+2860:BRAFdup Protein Fusion: in frame (KIAA1549-BRAF)		
EVT-P-0064872-T01-IM7-157	P-0064872-T01-IM7		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140489277	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4032+4347:KIAA1549_c.1141-1893:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}		

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	main_ref	clonality	fusion_status	domain	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0013777-T01-IM5-158	P-0013777-T01-IM5		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140481674	11	439	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			KIAA1549(NM_001164665) - BRAF (NM_004333) rearrangement: c.5248-2734:KIAA1549_c.1315-181:BRAFdup Protein fusion: in frame (KIAA1549-BRAF)	NA	
EVT-P-0040760-T01-IM6-159	P-0040760-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140481540	11	439	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			KIAA1549(NM_001164665) - BRAF (NM_004333) fusion: c.4929+2627:KIAA1549_c.1315-47:BRAFdup Protein fusion: in frame (KIAA1549-BRAF)	NA	
EVT-P-0014191-T01-IM6-160	P-0014191-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140492679	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			KIAA1549(NM_001164665) - BRAF (NM_004333) fusion: c.5248-320:KIAA1549_c.1140+1429:BRAFdup Protein fusion: in frame (KIAA1549-BRAF)	NA	
EVT-P-0014381-T01-IM6-161	P-0014381-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140492888	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			KIAA1549(NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-15 fused to BRAF exon 9 -18): c.4930-2082:KIAA1549_c.1140+1220:BRAFdup Protein fusion: in frame (KIAA1549-BRAF)	NA	
EVT-P-0028611-T01-IM6-162	P-0028611-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140482786	10	438	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			KIAA1549(NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused to BRAF exons 11-18): c.5248-4149:KIAA1549_c.1314+35:BRAFdup Protein fusion: in frame (KIAA1549-BRAF)	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	main_ref	donor_frame	maintained_domains	disrupted_domains	description	annotation	effect_on_fra
EVT-P-0028611-T02-IM6-168	P-0028611-T02-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140482786	10	438	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused to BRAF exons 11-18): c.5248-4149:KIAA1549_c.1314+	NA (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused to BRAF exons 11-18): c.5248-4149:KIAA1549_c.1314+	+35:BRAFdup
EVT-P-0043874-T01-IM6-169	P-0043874-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140490001	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5248-2907:KIAA1549_c.1141-2617:BRAFdup	{KIAA1549; BRAF} KIAA1549 NA (NM_001164665) - BRAF (NM_004333) fusion: c.5248-2907:KIAA1549_c.1141-2617:BRAFdup	
EVT-P-0069024-T04-IM7-170	P-0069024-T04-IM7		KIAA1549-BRAF	KIAA1549	unknown	three_prime	140492117	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5059:KIAA1549_c.1140+	{KIAA1549; BRAF} KIAA1549 NA (NM_001164665) - BRAF (NM_004333) fusion: c.5059:KIAA1549_c.1140+	+1991:BRAFdup
EVT-P-0029361-T01-IM6-171	P-0029361-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140489625	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) Fusion: c.4930-2667:KIAA1549_c.1141-2241:BRAFdup	{KIAA1549; BRAF} KIAA1549 NA (NM_001164665) - BRAF (NM_004333) Fusion: c.4930-2667:KIAA1549_c.1141-2241:BRAFdup	
EVT-P-0016557-T01-IM6-172	P-0016557-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140488576	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-15 fused in-frame to BRAF exons 9-18) : c.4929+2955:KIAA1549_c.1141-1192:BRAFdup	{KIAA1549; BRAF} KIAA1549 NA (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-15 fused in-frame to BRAF exons 9-18) : c.4929+2955:KIAA1549_c.1141-1192:BRAFdup	
EVT-P-0032982-T01-IM6-173	P-0032982-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140491506	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) Fusion (KIAA1549 exons 1-16 fused with BRAF exons 9-18) : c.5248-10:KIAA1549_c.1140+	{KIAA1549; BRAF} KIAA1549 NA (NM_001164665) - BRAF (NM_004333) Fusion (KIAA1549 exons 1-16 fused with BRAF exons 9-18) : c.5248-10:KIAA1549_c.1140+	+2602:BRAFdup

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_coding	breakpoint_4	breakpoint_5	main_ref	donor_frame	inframe	domains	disrupted_domains	description	annotation	effect_on_fra
EVT-P-0064310-T04-IM7-174	P-0064310-T04-IM7		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140493446	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5247+3560:KIAA1549_c.1140+662:BR		
EVT-P-0021127-T01-IM6-175	P-0021127-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140491741	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			Fusion: in frame {KIAA1549:KIAA1549 - BRAF (NM_001164665) (NM_004333) fusion (KIAA1549 exons 1-10 fused with BRAF exons 9-18) : c.4032+3996:KIAA1549_c.1140+2367:BR		
EVT-P-0017693-T01-IM6-176	P-0017693-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140489537	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			Fusion: in frame {KIAA1549:KIAA1549 - BRAF (NM_001164665) (NM_004333) fusion (c.1141-2153): c.4930-495:KIAA1549_c.1141-2153:BRADup		
EVT-P-0020125-T01-IM6-177	P-0020125-T01-IM6		KIAA1549-BRAF	KIAA1549	unknown	three_prime	140492175	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			Fusion: in frame {KIAA1549:KIAA1549 - BRAF (NM_001164665) (NM_004333) fusion (KIAA1549 exons 1-17 fused to BRAF exons 9-18): c.5260:KIAA1549_c.1140+1933:BRAD		
EVT-P-0045683-T02-IM6-178	P-0045683-T02-IM6		KIAA1549-BRAF	KIAA1549	unknown	three_prime	140492850	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			Fusion: mid-exon {KIAA1549:KIAA1549 - BRAF (NM_001164665) (NM_004333) fusion: c.5189:KIAA1549_c.1140+1258:BRAD		
EVT-P-0019127-T01-IM6-179	P-0019127-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140490657	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			Fusion: mid-exon {KIAA1549:KIAA1549 - BRAF (NM_001164665) (NM_004333) fusion (KIAA1549 exons 1-16 fused to BRAF exons 9-18): c.5247+2110:KIAA1549_c.1141-3273:BRADup		

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_pos	breakpoint_4	main_retained	donor_frame	transcript_maintained	domains	disrupted_domains	description_annotation	cancer_site	effect_on_function
EVT-P-0066022-T01-IM7-180	P-0066022-T01-IM7		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140488035	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5248-4074:KIAA1549_c.1141-651:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA	
EVT-P-0036680-T01-IM6-181	P-0036680-T01-IM6		KLHL12-BRAF	KLHL12	in-frame	three_prime	140493984	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KLHL12 (NM_021633) - BRAF (NM_004333) fusion (KLHL12 exons 1-4 fused to BRAF exons 9-18): t(1;7)(q32;q34)(chr1:g.202880970::chr7:		
EVT-P-0047833-T02-IM6-183	P-0047833-T02-IM6		LRBA-BRAF	LRBA	in-frame	three_prime	140493444	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		LRBA (NM_001199282) - BRAF (NM_004333) fusion: t(4;7)(q31;q34)(chr4:g.151502269::chr7:		
EVT-P-0006412-T01-IM5-184	P-0006412-T01-IM5		LUC7L2-BRAF	LUC7L2	unknown	five_prime	140501109	6	287	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		{LRBA:BRAF} LUC7L2 (NM_001244584) - BRAF (NM_004333) Rearrangement : c.1008-2090:LUC7L2_c.860+103:BRAFdel Antisense fusion MAD1L1 (NM_001013837) - BRAF (NM_004333) fusion: c.1417-9863:MAD1L1_c.1140+2359:BRAFdudup Protein Fusion: in frame {MAD1L1:BRAF}	NA	
EVT-P-0026158-T04-IM6-185	P-0026158-T04-IM6		MAD1L1-BRAF	MAD1L1	in-frame	three_prime	140491749	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MAD1L1 (NM_001013837) - BRAF (NM_004333) fusion: c.1417-9863:MAD1L1_c.1140+2359:BRAFdudup Protein Fusion: in frame {MAD1L1:BRAF}	NA	
EVT-P-0026158-T01-IM6-186	P-0026158-T01-IM6		MAD1L1-BRAF	MAD1L1	in-frame	three_prime	140491749	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MAD1L1 (NM_001013837) - BRAF (NM_004333) Rearrangement : c.1417-9863_c.1140+2359dup Protein Fusion: in frame {MAD1L1:BRAF}	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint	breakpoint	breakpoint	protein	protein	breakpoint	main_ref	donor	fract	main_ref	disrupted	description	annotation	effect_on
EVT-P-0006601-T03-IM6-187	P-0006601-T03-IM6		MINDY4-BRAF	MINDY4	in-frame	three_prime	140490696	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			FAM188B (NM_032222) - BRAF (NM_004333) Fusion (FAM188B exon5 fused with BRAF exon9) : c.664-366:FAM188B_c.1141-3312:BRAF Protein Fusion: in frame {FAM188B:BRAF}	NA	
EVT-P-0023220-T01-IM6-188	P-0023220-T01-IM6		MKLN1-BRAF	MKLN1	in-frame	three_prime	140496796	8	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			MKLN1 (NM_013255) - BRAF (NM_004333) rearrangement: c.98+4070:MKLN1_c.981-2529:BRAF Protein Fusion: in frame {MKLN1:BRAF}	NA	
EVT-P-0035393-T02-IM6-189	P-0035393-T02-IM6		MKRN1-BRAF	MKRN1	in-frame	three_prime	140488436	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			MKRN1 (NM_013446) - BRAF (NM_004333) fusion: c.771+833:MKRN1_c.1141-1052:BRAFdup Protein Fusion: in frame {MKRN1:BRAF}	NA	
EVT-P-0009024-T01-IM5-190	P-0009024-T01-IM5		MKRN1-BRAF	MKRN1	in-frame	three_prime	140493485	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			MKRN1 (NM_013446) - BRAF (NM_004333) rearrangement: c.772-184G:MKRN1_c.1140+623:BRAFdup Protein fusion: in frame (MKRN1-BRAF)	NA	
EVT-P-0008083-T03-IM6-191	P-0008083-T03-IM6		MKRN1-BRAF	MKRN1	out-of-frame	three_prime	140484511	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			MKRN1 (NM_013446) - BRAF (NM_004333) rearrangement: c.315-432:MKRN1_c.1178-1554:BRAFdup Protein Fusion: out of frame {MKRN1:BRAF}	NA	
EVT-P-0001036-T01-IM3-192	P-0001036-T01-IM3		MKRN1-BRAF	MKRN1	in-frame	three_prime	140482250	10	438	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			NA NA Protein fusion: in frame (MKRN1-BRAF)	NA	
EVT-P-0047135-T01-IM6-193	P-0047135-T01-IM6		MKRN1-BRAF	MKRN1	in-frame	three_prime	140492510	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			MKRN1 (NM_013446) - BRAF (NM_004333) fusion: c.772-972:MKRN1_c.1140+1598:BRAFdup Protein Fusion: in frame {MKRN1:BRAF}	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint	breakpoint_pos	breakpoint_pos	protein_position	domain_position	main_retained	donor_insertion	domain_deleted	disrupted_domains	sequence_annotation	transcript_effect_on_fusion
EVT-P-0043972-T03-IM6-200	P-0043972-T03-IM6		NRF1-BRAF	NRF1	in-frame	three_prime	140487712	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain	Fusion: NRF1(NM_005011) - BRAF(NM_004333) c.607-7410:NRF1_c.1141-328:BRAFinv Protein fusion: in frame {NRF1:BRAF}	NA
EVT-P-0065446-T01-IM7-201	P-0065446-T01-IM7		NRF1-BRAF	NRF1	in-frame	three_prime	140483810	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain	Fusion: NRF1(NM_005011) - BRAF(NM_004333) c.1349-10893:NRF1_c.1178-853:BRAFinv Protein fusion: in frame {NRF1:BRAF}	NA
EVT-P-0059002-T01-IM7-202	P-0059002-T01-IM7		OSBPL7-BRAF	OSBPL7	in-frame	three_prime	140491530	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain	Fusion: OSBPL7(NM_145798) - BRAF(NM_004333) t(7;17)(q34;q21.32)(chr7:g.140491530::chr17:g.11178853::) inv Protein fusion: in frame {OSBPL7:BRAF}	NA
EVT-P-0054958-T02-IM6-203	P-0054958-T02-IM6		PAK2-BRAF	PAK2	in-frame	three_prime	140486065	9	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain	Fusion: PAK2(NM_002577) - BRAF(NM_004333) rearrangement: t(3;7)(q29;q34)(chr3:g.196530579::chr7:g.11178853::) inv Protein fusion: in frame {PAK2:BRAF}	NA
EVT-P-0008789-T01-IM5-205	P-0008789-T01-IM5		PHTF2-BRAF	PHTF2	in-frame	three_prime	140489236	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain	Fusion: PHTF2(NM_001127358) - BRAF(NM_004333) rearrangement: c.2224-367:PHTF2_c.1141-1852:BRAFinv Protein fusion: in frame (PHTF2-BRAF)	NA
EVT-P-0044819-T01-IM6-206	P-0044819-T01-IM6		PLPP1-BRAF	PLPP1	out-of-frame	three_prime	140482123	11	439	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain	Fusion: PPAP2A(NM_176895) - BRAF(NM_004333) t(5;7)(q11.2;q34)(chr5:g.54789192::chr7:g.11178853::) inv Protein fusion: out of frame {PPAP2A:BRAF}	NA
EVT-P-0009694-T01-IM5-207	P-0009694-T01-IM5		PRKAR1B-BRAF	PRKAR1B	in-frame	three_prime	140491855	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain	Fusion: PRKAR1B(NM_001164758) -BRAFRearrangement : c.892-2551:PRKAR1B_c.1140+2253:BRAFdup Protein fusion: in frame (PRKAR1B-BRAF)	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_positives	protein_positives	main_retention_ratio	dondraft	unintended_domains	disrupted_domains	domains_annotation	position_effect_on_fra
EVT-P-0008567-T01-IM5-208	P-0008567-T01-IM5		PRKAR2B-BRAF	PRKAR2B	in-frame	three_prime	140486416	9	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich do-main		PRKAR2B (NM_002736) - BRAF (NM_004333) rearrangement: c.307+8063 PRKAR2B_c.1177+932:BR
EVT-P-0041181-T01-IM6-209	P-0041181-T01-IM6		PTPRZ1-BRAF	PTPRZ1	in-frame	three_prime	140492428	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich do-main		PTPRZ1 (NM_002851) - BRAF (NM_004333) fusion: c.5367+457 PTPRZ1_c.1140+1680:BR
EVT-P-0012884-T01-IM5-210	P-0012884-T01-IM5		PWWP2A-BRAF	PWWP2A	in-frame	three_prime	140495243	8	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich do-main		Fusion: in frame {PT-PRZ1:BRAF} PWWP2A (NM_001130864) - BRAF (NM_004333) Rearrangement : t(5;7)(q33;q34)(chr5:g.159539324::chr7:g.30030473::chr7:g.140493282::chr9:g.140485917::chr
EVT-P-0068260-T01-IM7-211	P-0068260-T01-IM7		RBMS3-BRAF	RBMS3	in-frame	three_prime	140482707	10	438	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich do-main		RBMS3 (NM_001003793) - BRAF (NM_004333) fusion: t(3;7)(p24.1;q34)(chr3:g.30030473::chr7:g.140485917::chr
EVT-P-0064764-T01-IM7-212	P-0064764-T01-IM7		RGS3-BRAF	RGS3	out-of-frame	three_prime	140493282	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich do-main		Fusion: in frame {RBMS3:BRAF} RGS3 (NM_144488) - BRAF (NM_004333) fusion: t(7;9)(q34;q32)(chr7:g.140493282::chr9:g.140485917::chr
EVT-P-0041308-T02-IM6-213	P-0041308-T02-IM6		RRBP1-BRAF	RRBP1	in-frame	three_prime	140485917	9	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich do-main		Fusion: out of frame {RGS3:BRAF} RRBP1 (NM_004587) - BRAF (NM_004333) fusion: t(7;20)(q34;p12.1)(chr7:g.140485917::chr
EVT-P-0007392-T01-IM5-214	P-0007392-T01-IM5		SCRIB-BRAF	SCRIB	in-frame	three_prime	140482150	11	439	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich do-main		Fusion: in frame {RRBP1:BRAF} SCRIB (NM_182706) - BRAF (NM_004333) rearrangement: t(7;8)(q34;q24.3)(chr7:g.140482150::chr8:g.140485917::chr

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	is_main_ref	is_fusion	is_inverted	is_deleted	is_disrupted	description	annotation	effect_on_fra
EVT-P-0029727-T01-IM6-222	P-0029727-T01-IM6		SND1-BRAF	SND1	in-frame	three_prime	140482257	10	438	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			SND1 (NM_014390) - BRAF (NM_004333) fusion: c.1152+12095:SND1_c.1314+564:BRAF	NA	
EVT-P-0068194-T01-IM7-223	P-0068194-T01-IM7		SND1-BRAF	SND1	in-frame	three_prime	140489175	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			SND1 (NM_014390) - BRAF (NM_004333) fusion: c.1152+17686:SND1_c.1141-1791:BRAF inv	NA	
EVT-P-0008140-T01-IM5-224	P-0008140-T01-IM5		SND1-BRAF	SND1	in-frame	three_prime	140489439	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			SND1 (NM_014390) - BRAF (NM_004333) fusion (SND1 exons 1-9 fused with BRAF exons 9-18): c.1039-3849:SND1_c.1141-2055:BRAF inv	NA	
EVT-P-0034168-T01-IM6-225	P-0034168-T01-IM6		SND1-BRAF	SND1	in-frame	three_prime	140489421	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			SND1 (NM_014390) - BRAF (NM_004333) fusion (SND1 exons 1-10 fused to BRAF exons 9-18): c.1152+6972:SND1_c.1141-2037:BRAF inv	NA	
EVT-P-0025075-T01-IM6-226	P-0025075-T01-IM6		SND1-BRAF	SND1	in-frame	three_prime	140490546	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			SND1 (NM_014390) - BRAF (NM_004333) fusion (SND1 exons 1-9 with BRAF exons 9-18): c.1039-2471:SND1_c.1141-3162:BRAF inv	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	is_main_ref	is_disrupted	domain_1	domain_2	domain_3	domain_4	disrupted_domains	description	annotation	effect_on_fra
EVT-P-0002071-T02-IM5-227	P-0002071-T02-IM5		SND1-BRAF	SND1	in-frame	three_prime	140489292	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain					SND1 (NM_014390) - BRAF (NM_004333) fusion (SND1 exons 1-10 fused with BRAF exons 9-18) : c.1152+18606:SNV Protein fusion: in frame (SND1-BRAF) c.1528-8436:SNV Protein fusion: in frame (BRAF-SND1)	NA	1141-
EVT-P-0002197-T01-IM3-228	P-0002197-T01-IM3		SND1-BRAF	SND1	in-frame	five_prime	140488303	9	381	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain					c.1528-8436:SNV Protein fusion: in frame (BRAF-SND1)	NA	1141-
EVT-P-0007533-T01-IM5-229	P-0007533-T01-IM5		SND1-BRAF	SND1	in-frame	three_prime	140482226	10	438	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain					SND1 (NM_014390) - BRAF (NM_004333) fusion (SND1 exons 1-10 with BRAF exons 11-18) : c.1152+31738:SNV Protein fusion: in frame (SND1-BRAF)	NA	1314+595:BRAF
EVT-P-0036567-T01-IM6-230	P-0036567-T01-IM6		SND1-BRAF	SND1	in-frame	three_prime	140493552	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain					SND1 (NM_014390) - BRAF (NM_004333) Fusion (SND1 exon10 fused with BRAF exon9) : c.1153-22013:SNV Protein Fusion: in frame {SND1:BRAF}	NA	1141-
EVT-P-0034845-T01-IM6-231	P-0034845-T01-IM6		SND1-BRAF	SND1	in-frame	three_prime	140489241	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain					SND1 (NM_014390) - BRAF (NM_004333) rearrangement: c.1527+3658:SNV Protein Fusion: in frame {SND1:BRAF}	NA	1141-
EVT-P-0054364-T01-IM6-232	P-0054364-T01-IM6		SNX8-BRAF	SNX8	in-frame	three_prime	140488730	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain					SNX8 (NM_013321) - BRAF (NM_004333) fusion: c.1284+565:SNV Protein Fusion: in frame {SNX8:BRAF}	NA	1141-

[illegible]

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	inframe	main_ref	disrupted	description	annotation	effect_on_fra
EVT-P-0027762-T01-IM6-250	P-0027762-T01-IM6		ZC3HAV1-BRAF	ZC3HAV1	in-frame	three_prime	140485946	9	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		ZC3HAV1 (NM_020119) - BRAF (NM_004333) rearrangement: c.698-431:ZC3HAV1_c.1177+1402:BRAFdup	
EVT-P-0008920-T01-IM5-251	P-0008920-T01-IM5		ZNF207-BRAF	ZNF207	in-frame	three_prime	140485847	9	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		Fusion: in frame {ZC3HAV1:BRAF} ZNF207 (NM_001098507) - BRAF (NM_004333) rearrangement: t(7;17)(q34;q11.2)(chr7:g.140485847::chr	

3 Figures

BRAF domain-retention track

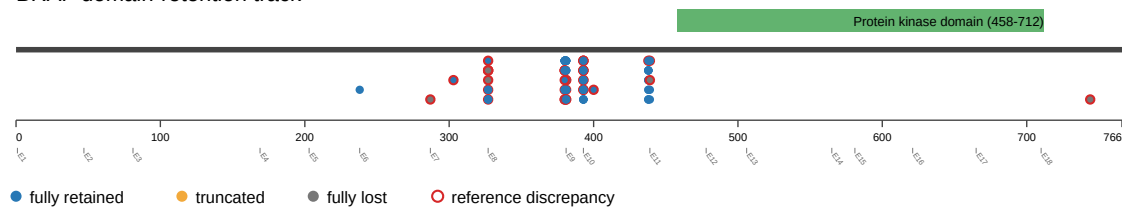
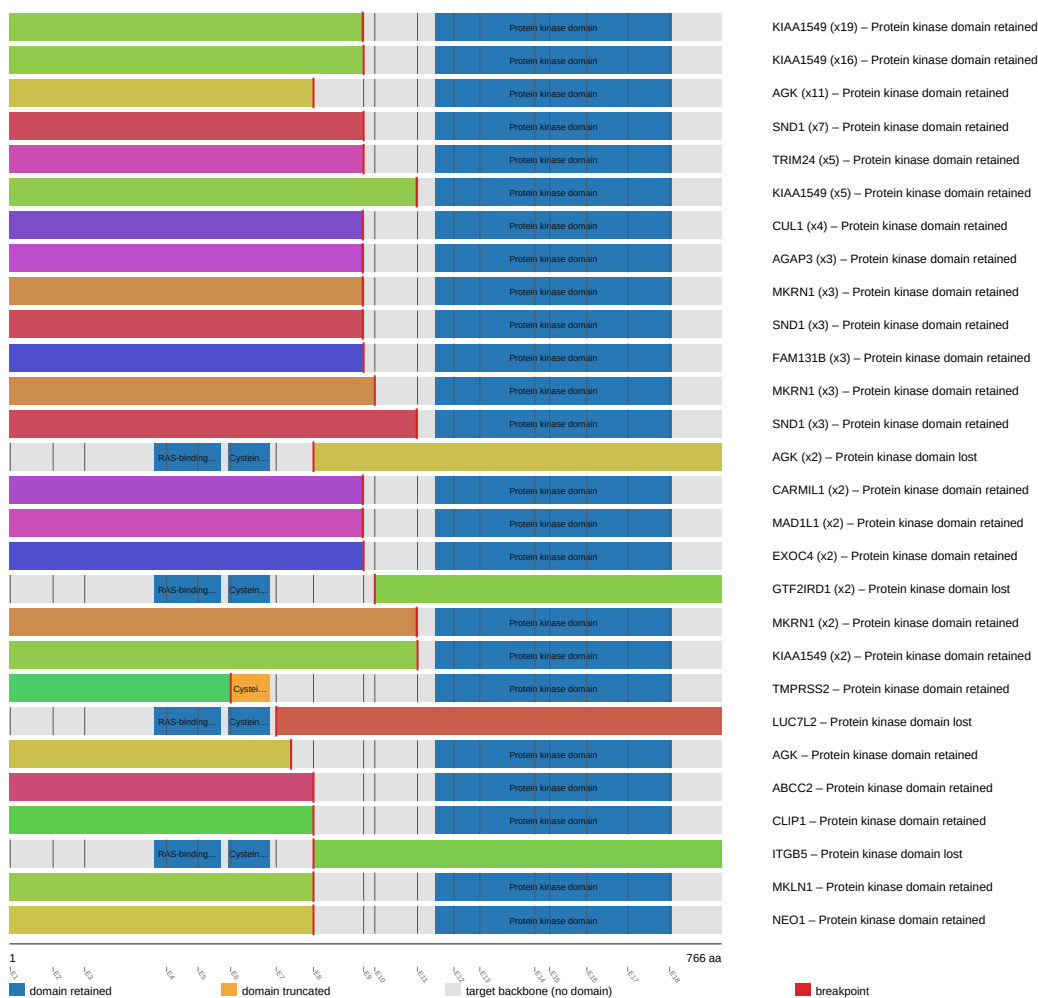


Figure 1: domain retention outliers

BRAF fusion-transcript schematic

partner block → breakpoint → retained BRAF portion (domain-colored, exon ticks)



Showing the top 28 of 99 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

Figure 2: fusion schematic

BRAF intragenic-deletion schematic

retained N-terminal block → deleted span (plain connector) → resumed C-terminal block

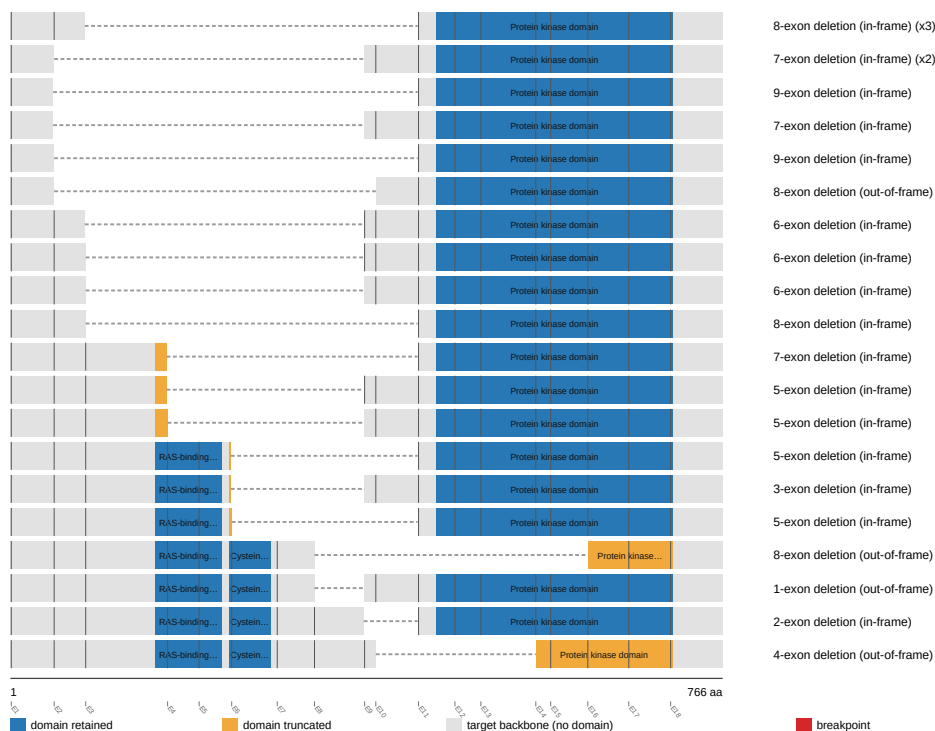


Figure 3: intragenic deletion schematic

Reference vs msk_impact_50k_2026



Figure 4: reference comparison